Zhenghao Zeng

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Research Interests

Causal inference, nonparametric statistics, high-dimensional statistics

Academic Position

Stanford University Stanford, CA
Postdoc Scholar at Graduate School of Business 09/2025 – present
Mentors: Prof. Stefan Wager, Ramesh Johari and Guido Imbens

Education

Carnegie Mellon University Pittsburgh, PA
Ph.D. in Statistics 08/2020 – 08/2025
Advisor: Prof. Edward H. Kennedy

Carnegie Mellon University Pittsburgh, PA
M.S. in Statistics 08/2020 – 05/2021

University of Science and Technology of China Hefei, Anhui
B.S. in Statistics 08/2016 – 06/2020

Honors and Awards

Student Paper Award, Statistical Learning and Data Science Section, ASA 2024
ENAR Distinguished Student Paper Award (International Biometric Society ENAR Spring Meeting) 2023
Guo Moruo Scholarship (summa cum laude at USTC) 2019
National Scholarship of China (USTC) 2018

Publications

Zeng, Z., Arbour, D., Feller, A., Dasgupta, I., Sinha, A., and Kennedy, E. H. Handling Missing Responses under Cluster Dependence with Applications to Language Model Evaluation. *Advances in Neural Information Processing Systems*, 2025.

Levis, A. W., Bonvini, M., **Zeng, Z.**, Keele, L., and Kennedy, E. H. (2023). Covariate-assisted bounds on causal effects with instrumental variables. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, qkaf028.

Du, J. H., **Zeng, Z.**, Kennedy, E. H., Wasserman, L., and Roeder, K. (2025). Causal inference for genomic data with multiple heterogeneous outcomes. *Journal of the American Statistical Association*, (just-accepted), 1-24.

Zeng, Z., Kennedy, E. H., Bodnar, L. M., and Naimi, A. I. (2023). Efficient generalization and transportation. arXiv preprint arXiv:2302.00092. Accepted by *Statistical Science*.

Zeng, Z., Arbour, D., Feller, A., Addanki, R., Rossi, R., Sinha, R., and Kennedy, E.H. (2024). Continuous Treatment Effects with Surrogate Outcomes. *International Conference on Machine Learning*, 2024.

Zeng, Z., Gu, Y., and Xu, G. (2023). A Tensor-EM Method for Large-Scale Latent Class Analysis with Binary Responses. *Psychometrika*, 88(2), 580-612.

He, Y., Meng, B., **Zeng, Z.**, and Xu, G. (2021). On the phase transition of Wilks' phenomenon. *Biometrika*, 108(3), 741-748.

Preprints

Zeng, Z., Adjaho, C., Bucarey, A., Qin, C., Zhang, R., Hoban, P., Johari, R. and Wager, S. (2026). Sequentially-Rerandomized Switchback Experiments. arXiv preprint arXiv:2604.02489.

Zeng, Z., Levis, A.W., Lee, J.H., Kennedy, E.H., and Keele, L. (2025). Nonparametric Estimation of Local Treatment Effects with Continuous Instruments. arXiv preprint arXiv:2504.03063.

Zeng, Z., Balakrishnan, S., Han, Y., and Kennedy, E. H. (2024). Causal Inference with High-dimensional Discrete Covariates. arXiv preprint arXiv:2405.00118.

Bonvini, M., **Zeng, Z.**, Yu, M., Kennedy, E. H., and Keele, L. (2023). Flexibly Estimating and Interpreting Heterogeneous Treatment Effects of Laparoscopic Surgery for Cholecystitis Patients. arXiv preprint arXiv:2311.04359.

Experience

Adobe Research — San Jose, CA
Research Intern — 05/2024 – 08/2024
Mentors: David Arbour, Ishita Dasgupta, Atanu Sinha and Avi Feller (Berkeley)
Handling missing responses under cluster dependence with applications to language model evaluation.

Adobe Research — San Jose, CA
Research Intern — 05/2023 – 08/2023
Mentors: David Arbour and Avi Feller (Berkeley)
Continuous treatment effects with surrogate outcomes.

University of Michigan, Ann Arbor — Ann Arbor, MI
Research Assistant — 06/2019 – 09/2019
Mentor: Prof. Gongjun Xu
Large-scale latent class analysis and high-dimensional testing.

Teaching

Teaching assistant, Carnegie Mellon University — 08/2020–05/2022
36-225 Introduction to Probability

36-401 Modern Regression
36-402 Advanced Methods for Data Analysis (×2)

Teaching assistant, USTC 09/2019 – 05/2020
Single-variable Calculus
Regression Analysis

Talks

Econometrics Lunch Stanford, CA
Causal inference with high-dimensional discrete covariates 01/2025

International Conference on Computational & Methodological Statistics (CMStatistics) Virtual
Continuous treatment effects with surrogate outcomes 12/2024

Joint Statistical Meetings (JSM) Portland, OR
Causal inference with high-dimensional discrete covariates 08/2024

American Causal Inference Conference (ACIC) Seattle, WA
Causal inference with high-dimensional discrete covariates 05/2024

Causal Inference and Missing Data Group at Inria Virtual
Efficient generalization and transportation 02/2024

American Causal Inference Conference (ACIC) Austin, TX
Causal inference with high-dimensional discrete covariates 05/2023

International Biometric Society ENAR Spring Meeting Nashville, TN
Efficient generalization and transportation 03/2023

Academic Service

Reviewer
Annals of Statistics (1)
American Journal of Epidemiology (1)
Biometrika (5)
Biostatistics (2)
Electronic Journal of Statistics (5)
Journal of American Statistical Association (10)
Journal of Causal Inference (5)
Journal of Machine Learning Research (1)
Journal of the Royal Statistical Society: Series B (2)
NeurIPS (5)
Stat (1)
Statistics in Medicine (1)
Statistica Sinica (1)
The American Statistician (1)

Organizer

CMU Causal Inference Reading Group (2024-2025)